

Resolution of Alarm ALARM_1901 on Cisco-PE-Delhi-NCR-33

Node: Cisco-PE-Delhi-NCR-33 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_1901

Description

This SOP is for resolving the ALARM_1901 (Warning) on Cisco-PE-Delhi-NCR-33, a Provider Edge Router from Cisco.

Potential Root Causes

Cause	Probability
Router hardware failure or misconfiguration	Medium
Link or interface failure	Medium
Routing protocol misconfiguration	Low

Diagnostic Commands

```
show controller
show ip int brief
show ip bgp summary
show ip bgp neighbor
debug ip bgp < neighbor_IP>
```

Resolution Steps

- Step 1: {'Step': 'Verify interface status and configuration', 'Action': 'Run show ip int brief and show controller commands'}
- Step 2: {'Step': 'Check link to neighboring routers', 'Action': 'Run ping and traceroute commands'}
- Step 3: {'Step': 'Verify BGP neighbor status', 'Action': 'Run show ip bgp neighbor and debug ip bgp commands'}
- Step 4: {'Step': 'Check for routing protocol misconfiguration', 'Action': 'Run show run and show ip bgp summary commands'}
- Step 5: {'Step': 'Replace failed hardware or reconfigure interfaces', 'Action': 'Perform necessary hardware replacement or reconfiguration'}

Post-Check Verification

```
show ip int brief
```

```
show ip bgp summary  
show ip bgp neighbor  
ping and traceroute
```